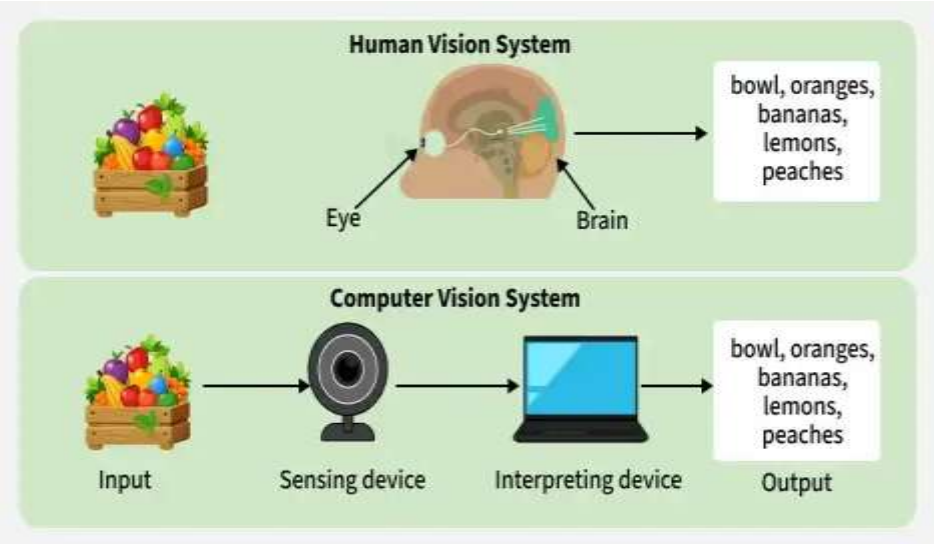
Day – 20 Introduction to Computer Vision

Computer Vision is a field of artificial intelligence that enables machines to interpret and understand visual information from images and videos. It uses image processing techniques and deep learning models to detect objects, recognize patterns and extract meaningful insights from visual data.

- Enables machines to analyze and understand images and videos.
- Helps identify objects, faces, text and other visual patterns.
- Supports tasks such as image classification, object detection and facial recognition.
- Widely used in healthcare, automotive, security and entertainment industries.
- Allows AI systems to make decisions based on visual information.



Main Components of Computer Vision:

Computer Vision relies on several techniques that help machines analyze and understand visual data effectively.

1. **Image Processing:** Enhances images by removing noise, improving contrast and adjusting brightness or colors.
2. **Object Detection:** Identifies and locates specific objects within an image or video.
3. **Image Classification:** Assigns an image to a predefined category or label.
4. **Feature Extraction:** Identifies important patterns such as shapes, colors, edges and textures for further analysis.

Working of Computer Vision

Computer Vision follows a series of steps to capture, process and analyze visual data, enabling machines to understand and make decisions based on images or videos.

1. Image Acquisition

- Images or videos are captured using cameras, sensors, or other devices.
- The quality and type of data influence the accuracy of the system.

2. Preprocessing

- Raw images are cleaned and enhanced before analysis.
- Common tasks include noise removal, brightness adjustment and image sharpening.

3. Feature Detection

- Important features such as edges, shapes, textures and patterns are identified.
- Helps the system focus on relevant information in the image.

4. Pattern Recognition

- Detected features are compared with learned patterns using machine learning models.
- Enables object recognition, image classification and scene understanding.

5. Decision Making

- The system uses the identified patterns to make predictions or take actions.
- Examples include recognizing faces, detecting objects or identifying traffic signs.

Tasks of Computer Vision:

Computer Vision performs a variety of tasks that enable machines to understand, analyze and interpret visual information from images and videos.

1. Object detection identifies and locates objects within an image or video by determining their positions, often using bounding boxes around each detected object.
2. Face recognition is used to recognize and verify individuals based on their facial features.
3. Image classification assigns an image to a predefined category or label based on its content.
4. Image segmentation divides an image into smaller meaningful regions for detailed analysis.
5. Optical Character Recognition (OCR) extracts and recognizes text from images, scanned documents and signboards.

6. Pose estimation identifies and tracks the position and movement of different parts of the human body.
7. Medical image analysis helps detect diseases and abnormalities from medical images such as X-rays, CT scans and MRI scans.

Applications

- Used in healthcare to analyze medical images and assist in disease detection.
- Helps self-driving vehicles recognize roads, traffic signs and obstacles.
- Improves security through surveillance and face recognition systems.
- Supports crop monitoring and pest detection in agriculture.
- Enables quality inspection and defect detection in manufacturing.

Advantages

- Processes large volumes of images and videos quickly, making it suitable for real-time applications.
- Delivers consistent results without fatigue, even for repetitive tasks.
- Can handle and analyze large scale visual data efficiently.
- Provides high accuracy in tasks such as object detection, image classification, and medical image analysis.
- Reduces manual effort by automating visual inspection and monitoring tasks.

Limitations

- Performance can be affected by poor lighting conditions, shadows or glare.
- Objects that are partially hidden can be difficult to detect accurately.
- Complex backgrounds and visual noise may reduce accuracy.
- Requires large amounts of high quality labeled data for effective training.
- Performance may vary when images differ significantly from the training data.